

Electrical Engineering Student

Nir Cohen

Profile

Third-year Electrical Engineering student, interested in digital design, hardware verification, embedded systems, and signal processing.

Comfortable with Python, C, C++, and MATLAB, with hands-on experience building real-time and hardware-software systems such as a piano note recognition pipeline (Arduino, signal processing) and a PID controller for an inverted-pendulum car (C++).

Curious, detail-oriented, and motivated by understanding how complex systems behave and ensuring they operate reliably.

Collaborative and always looking to learn from teammates and grow as an engineer.

Education

Tel Aviv University — B.Sc. Electrical Engineering | GPA: 82 (03/2023 – 2027)

The Open University — Academic Pathway Program | GPA: 85 (09/2020 – 03/2023)

Projects

Music Source Separation via NMF (Python, Signal Processing)

- Implemented Non-negative Matrix Factorization (NMF) from scratch to decompose audio signals into individual instrumental components.
- Developed an automated pipeline for Spectrogram pre-processing, utilizing STFT (Short-Time Fourier Transform) and specialized activation/basis matrix initialization.

Real-Time Piano Note Recognition System (C, Arduino, Signal Processing)

- Designed and implemented note recognition using harmonic frequency analysis, leveraging hardware-level features for real-time performance.
- Employed a dynamic scoring algorithm to determine the most likely note based on multiple frequency harmonics.
- Achieved top grade (**100**) in Hardware Lab.

PID Controller Implementation (C++)

- Implemented PID controller in C++ for precise control of an inverted pendulum car.

Academic Highlights

- Data Structures & Algorithms – **98**
- Digital Logic – **86**
- Signals & Systems – **86**
- Linear Algebra – **94**

Experience

Lab Technician | IDF Air Force | Hatzor Airbase (04/2020 - 12/22)

- Maintained and calibrated Scanning Electron Microscope and Color Spectrometry Equipment for propulsion system testing. Performed laboratory testing
- Designed an optimization fixture, reducing test times and waste; Received an **award** for innovation

Skills

Programming: Python (4+ yrs), C, C++, MATLAB

Embedded & Hardware: Arduino (5+ yrs), Raspberry Pi

Languages: Hebrew (Native), English (Fluent)